

##contact: Bo Meng
##bomeng@cau.edu.cn

Jena_time_NBE.csv – This dataset contains climate-related data along with the partitioned components of biodiversity effects on ecosystem functioning and variability over time within the main JENA experiment.

dBEF_time_NBE.csv – This dataset contains climate-related data along with the partitioned components of biodiversity effects on ecosystem functioning and variability over time within the dBEF experiment.

jena_detrend_SPEI.csv – This dataset contains the detrended biodiversity effects on ecosystem variability and the SPEI for each moving time window.

Variable Descriptions for "Jena_time_NBE.csv", "dBEF_time_NBE.csv", and "jena_detrend_SPEI.csv",

1. Design & Temporal Variables

block: The experimental block.

plot: The experimental plot.

Richness: Sown species richness.

n: Residual species richness.

Order: The chronological order of the moving time window (e.g., 1 denotes the 2003–2007 period, 2 denotes the 2004–2008 period).

2. Environmental Variables

T_mean: Mean annual temperature (°C) within the observed time window.

P_mean: Mean annual precipitation (mm) within the observed time window.

SPEI_mean: Mean Standardised Precipitation-Evapotranspiration Index (SPEI) within the observed time window.

Driest: The minimum SPEI (driest year) recorded within the time window.

3. Community Biomass & Structure

miu_mix: Mean biomass (g m⁻²) of the observed mixture community.

miu_mono: Expected mean biomass (g m⁻²) of the community in the absence of interspecific interactions (based on monocultures).

evenness_mix: Species evenness of the observed mixture community.

evenness_mono: Expected species evenness of the community in the absence of interspecific interactions (based on monocultures).

CorS_mono: Mean pairwise correlation among species in the expected community without interactions (based on monocultures).

cov.mono: Covariance between the column means and the coefficients of variation (CV) across species in the expected community without interactions (based on monocultures).

4. Biodiversity Effects on Ecosystem Functioning

NBE.F: Net Biodiversity Effect (NBE) on ecosystem functioning (community productivity).

SE: Selection Effect (SE) on ecosystem functioning (community productivity).

CE: Complementarity Effect (CE) on ecosystem functioning (community productivity).

5. Temporal Variability & Synchrony

CV.mix: Temporal variability (Coefficient of Variation) of the observed mixture community.

CV.mono: Expected temporal variability of the community in the absence of interspecific interactions (based on monocultures).

CV.avg: Unweighted average of individual species' temporal variability (baseline variability).

CVS.mix: Biomass-weighted species temporal variability in the observed mixture community.

CVS.mono: Expected biomass-weighted species temporal variability in the community without interactions (based on monocultures).

phi.mix: Species synchrony within the observed mixture community.

phi.mono: Expected species synchrony within the community in the absence of interspecific interactions (based on monocultures).

6. Interaction Effects on Stability Components

NBE: Net Biodiversity Effect on community temporal variability.

NIIE: Interspecific interaction-independent effect of biodiversity on community variability.

NIDE: Interspecific interaction-dependent effect of biodiversity on community variability.

NIIE.SpVar: Interspecific interaction-independent effect of biodiversity on species variability.

NIIE.Syn: Interspecific interaction-independent effect of biodiversity on species synchrony.

NIDE.SpVar: Interspecific interaction-dependent effect of biodiversity on species variability.

NIDE.Syn: Interspecific interaction-dependent effect of biodiversity on species synchrony.

AE.SpVar: Average effect of interspecific interactions on species variability.

SE.SpVar: Selection effect of interspecific interactions on species variability.

AE.Syn: Average effect of interspecific interactions on species synchrony.

SE.Syn: Selection effect of interspecific interactions on species synchrony.